STRUCTURE & FUNCTIONS OF HEMOGLOBIN

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BIOCHEMISTRY

HEME PROTEINS

- These are a group of specialized proteins that contain heme and globin.
- Heme is the prosthetic part and globin is the protein part.
- 97% is the globin part and the rest 3% is the heme part.

GLOBULAR HEME PROTEINS

- Role of heme group is dictated by the environment.

- Examples:
  a. Cytochromes
  b. Catalase
  c. Hemoglobin
  d. Myoglobin

HEMOGLOBIN

- Hemoglobin is the protein that carries oxygen from the lungs to the tissues and carries carbon dioxide from the tissues back to the lungs..

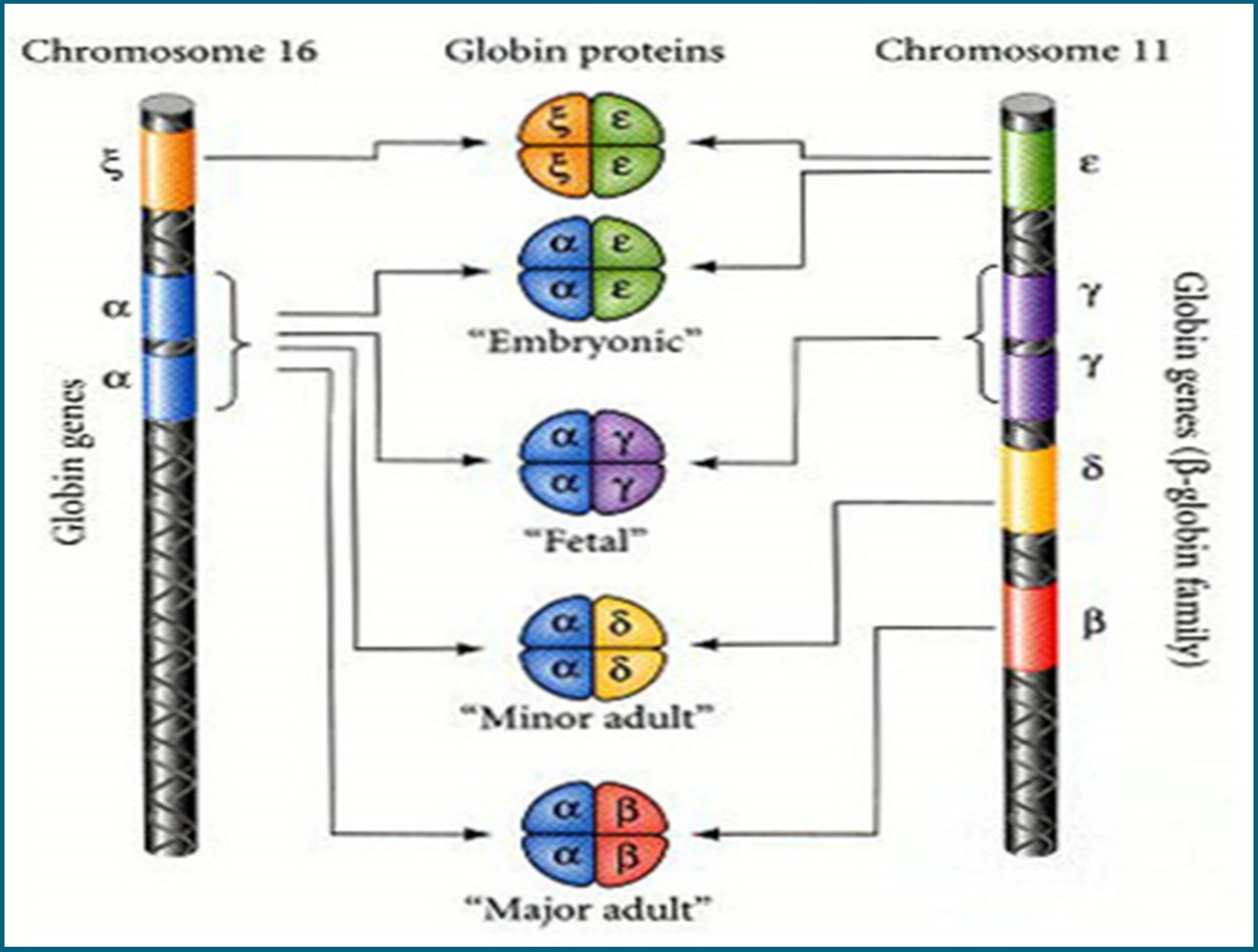
- The oxygen-carrying protein hemoglobin was discovered in 1840.

- Hemoglobin's reversible oxygenation was described a few years later.

- In 1959 Max Perutz determined the molecular structure of hemoglobin by X-ray crystallography. This work resulted in his sharing with John Kendrew the 1962 Nobel Prize in Chemistry.

STRUCTURE OF HEMOGLOBIN

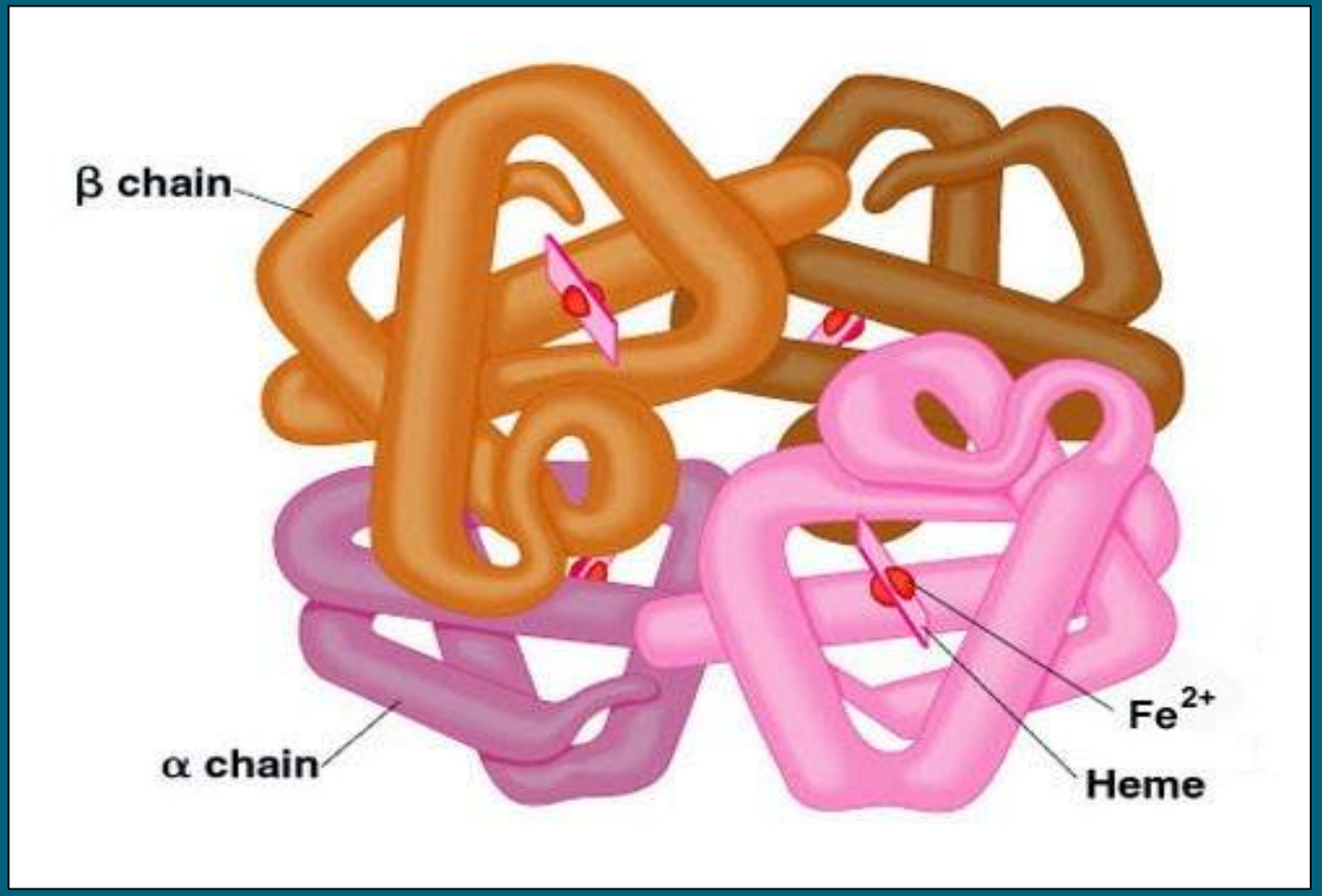
- Hemoglobin molecule consists of four polypeptide chains:
- Two alpha chains, each with 141 amino acids and
- Two beta chains, each with 146 amino acids.

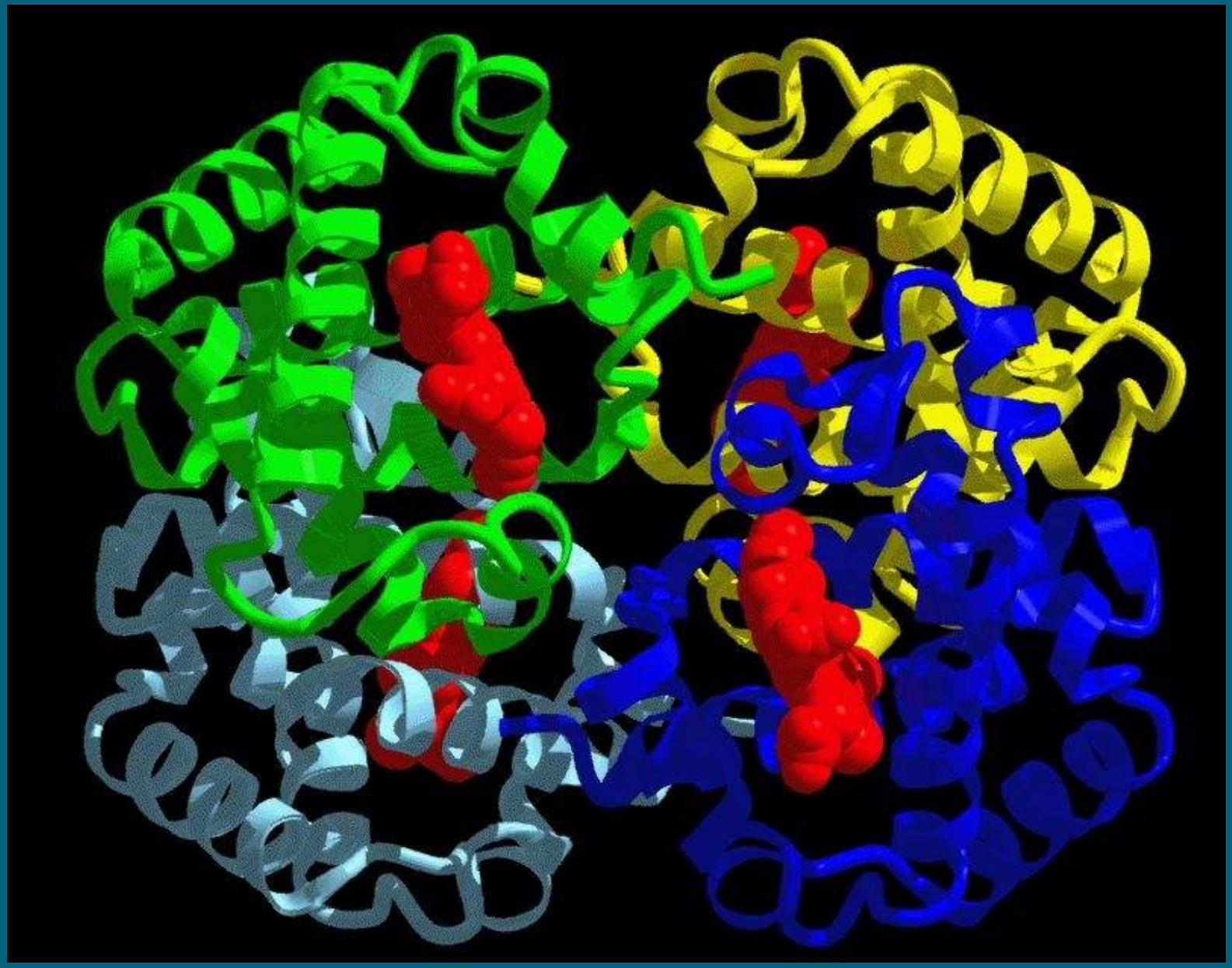

Chromosome 16
Globin proteins
Chromosome 11
ξ
α
α
Globin genes
ε
γ
γ
δ
β
Globin genes (β-globin family)
"Embryonic"
"Fetal"
"Minor adult"
"Major adult"

- Both the α and β globin chains contain primarily α helix secondary structure with no β sheets.

- Each α or β globin chain folds into 8 α - helical segments (A-H) which, in turn, fold to form globular tertiary structures.

- The folded helices form a pocket that holds the working part of each chain, the heme.


β chain
α chain
Fe2+
Heme



- The alpha and beta subunits of the globin chains exist in two dimers which are bonded together strongly.

- Thus Hb is a tetramer composed of two identical dimers $(\alpha\beta)_1$ $(\alpha\beta)_2$.

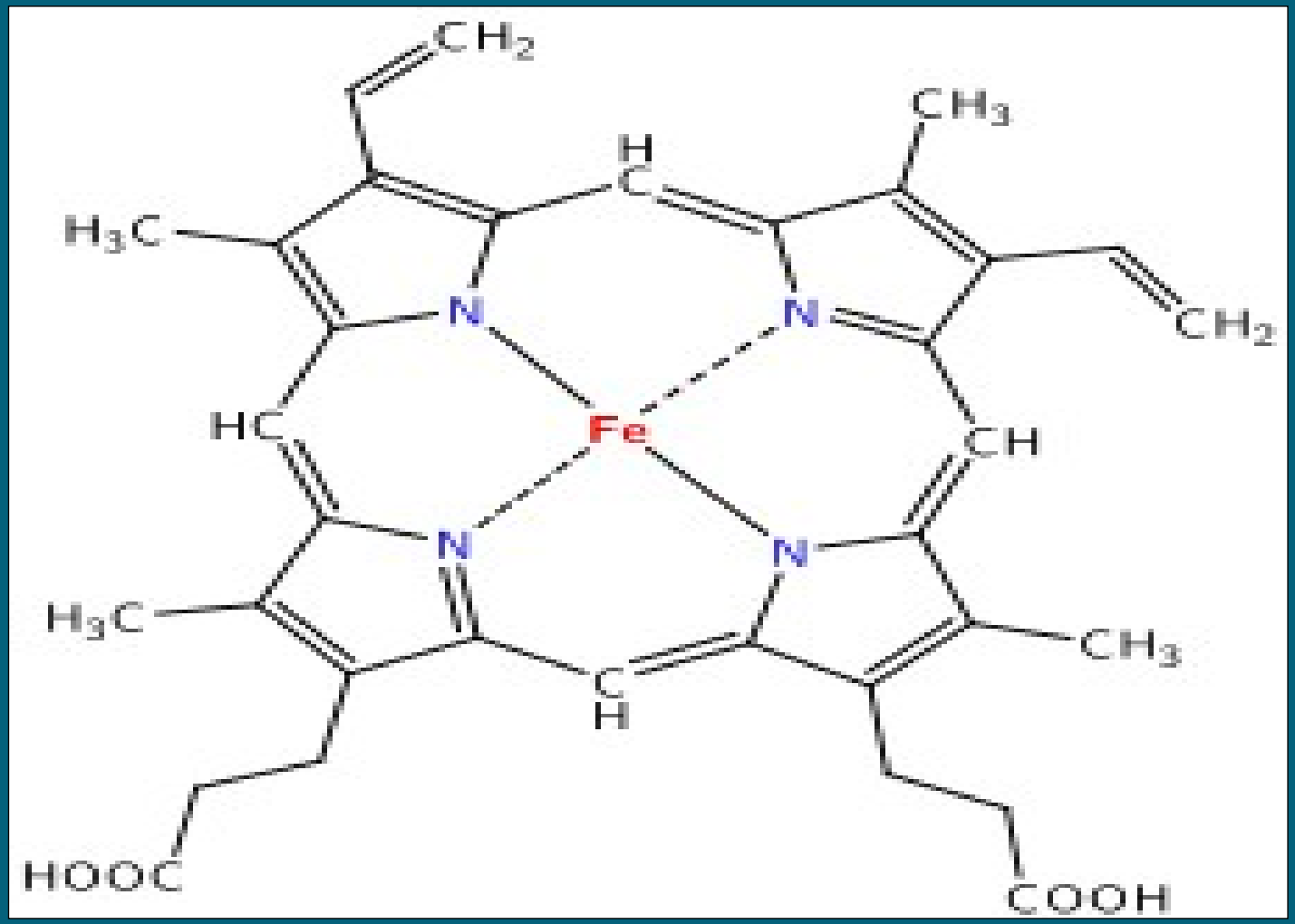
Quaternary structure of Hb

- Polypeptides of α and β chain are held together by hydrophobic interactions.
- Polypeptides between αβ dimers are held by ionic and hydrogen bonds.

PORPHYRIN METABOLISM

- Porphyrins are cyclic compounds.
- They bind metal ions, mostly Fe^{2+} or Fe^{3+}
- The most prevalent metalloporphyrin in humans is Heme.
- Heme is the prosthetic group for myoglobin, hemoglobin , cytochromes, catalase and tryptophan pyrrolase.

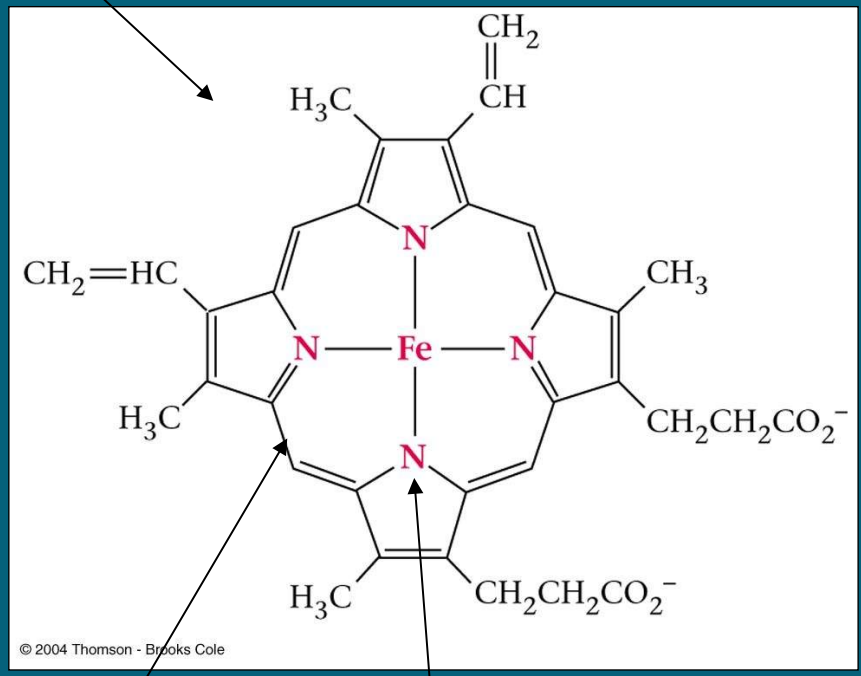
- Heme consists of one ferrous ion in the center of a tetrapyrrole ring of protoporphyrin IX.


CH2
CH3
H
H3C
N
N
CH2
HC
Fe
CH
N
N
H3C
CH3
C
H
HOOC
COOH

Heme structure

Heme is a metaloporphyrine (cyclic tetrapyrrole)

- Heme contains:
  - conjugated system of double bonds → ***red colour***
  - 4 nitrogen (N) atoms
  - 1 iron cation (Fe^{2+})
    → bound in the middle of tetrapyrrole skelet by **coordination covalent bonds**



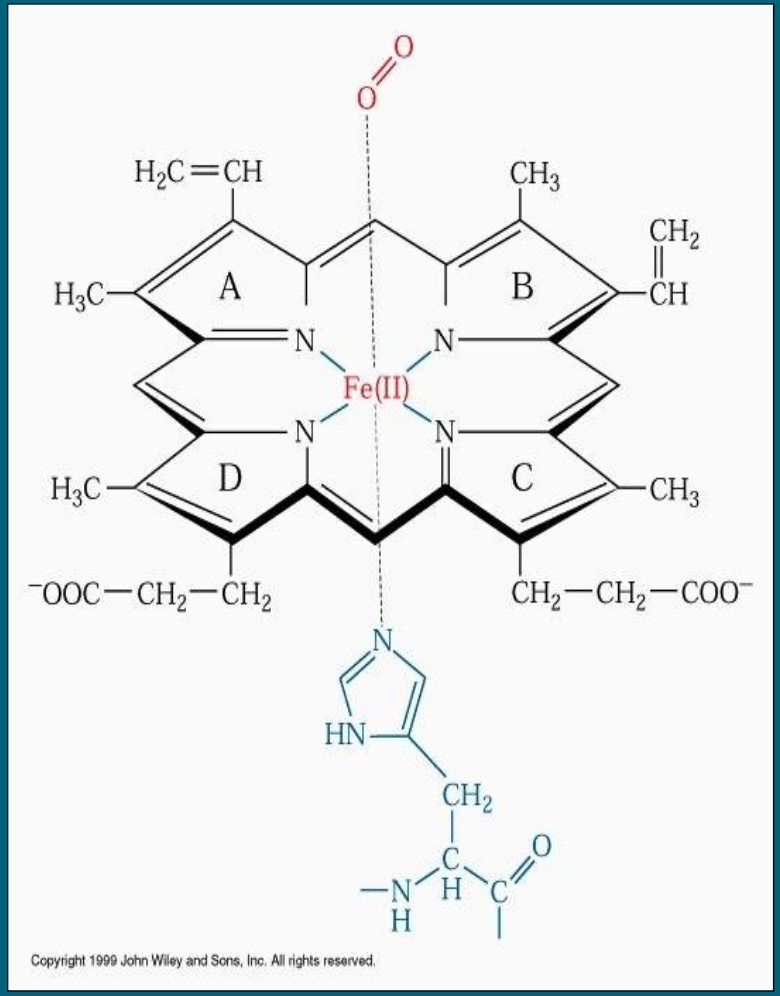
methine bridge pyrrole ring

Properties of iron in heme

- **Coordination number of iron in heme = 6**

6 bonds:

- 4x pyrrole ring (A,B,C,D)
- 1x link to a protein
- 1x link to an oxygen



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Types in humans

In the embryo:

- Gower 1 ($\zeta_2\varepsilon_2$)
- Gower 2 ($\alpha_2\varepsilon_2$)
- Hemoglobin Portland ($\zeta_2\gamma_2$)

In the fetus:

- Hemoglobin F ($\alpha_2\gamma_2$)

Types of Hb

Normal	HbA (97%)
	HbA2 (2%)
	HbF (1%)
	HbA_{1c}
abnormal	Carboxy Hb
	Met Hb
	Sulf Hb

In adults:

- Hemoglobin A ($\alpha_2\beta_2$) - The most common with a normal amount over 95%
- Hemoglobin A_2 ($\alpha_2\delta_2$) - δ chain synthesis begins late in the third trimester and in adults, it has a normal range of 1.5-3.5%
- Hemoglobin F ($\alpha_2\gamma_2$) - In adults Hemoglobin F is restricted to a limited population of red cells called F-cells. However, the level of Hb F can be elevated in persons with sickle-cell disease.

- Hemoglobin C ($\alpha_2\beta^C_2$) - Another variant due to a variation in the β-chain gene. This variant causes a mild chronic hemolytic anemia.

- Hemoglobin E ($\alpha_2\beta^E_2$) - Another variant due to a variation in the β-chain gene. This variant causes a mild chronic hemolytic anemia.

Variant forms which cause disease:

- Hemoglobin H (β_4) - A variant form of hemoglobin, formed by a tetramer of β chains, which may be present in variants of α thalassemia.

- Hemoglobin Barts (γ_4) - A variant form of hemoglobin, formed by a tetramer of γ chains, which may be present in variants of α thalassemia.

- Hemoglobin S ($\alpha_2\beta^S_2$) - A variant form of hemoglobin found in people with sickle cell disease. There is a variation in the β-chain gene, causing a change in the properties of hemoglobin which results in sickling of red blood cells.

- Hemoglobin AS - A heterozygous form causing Sickle cell trait with one adult gene and one sickle cell disease gene

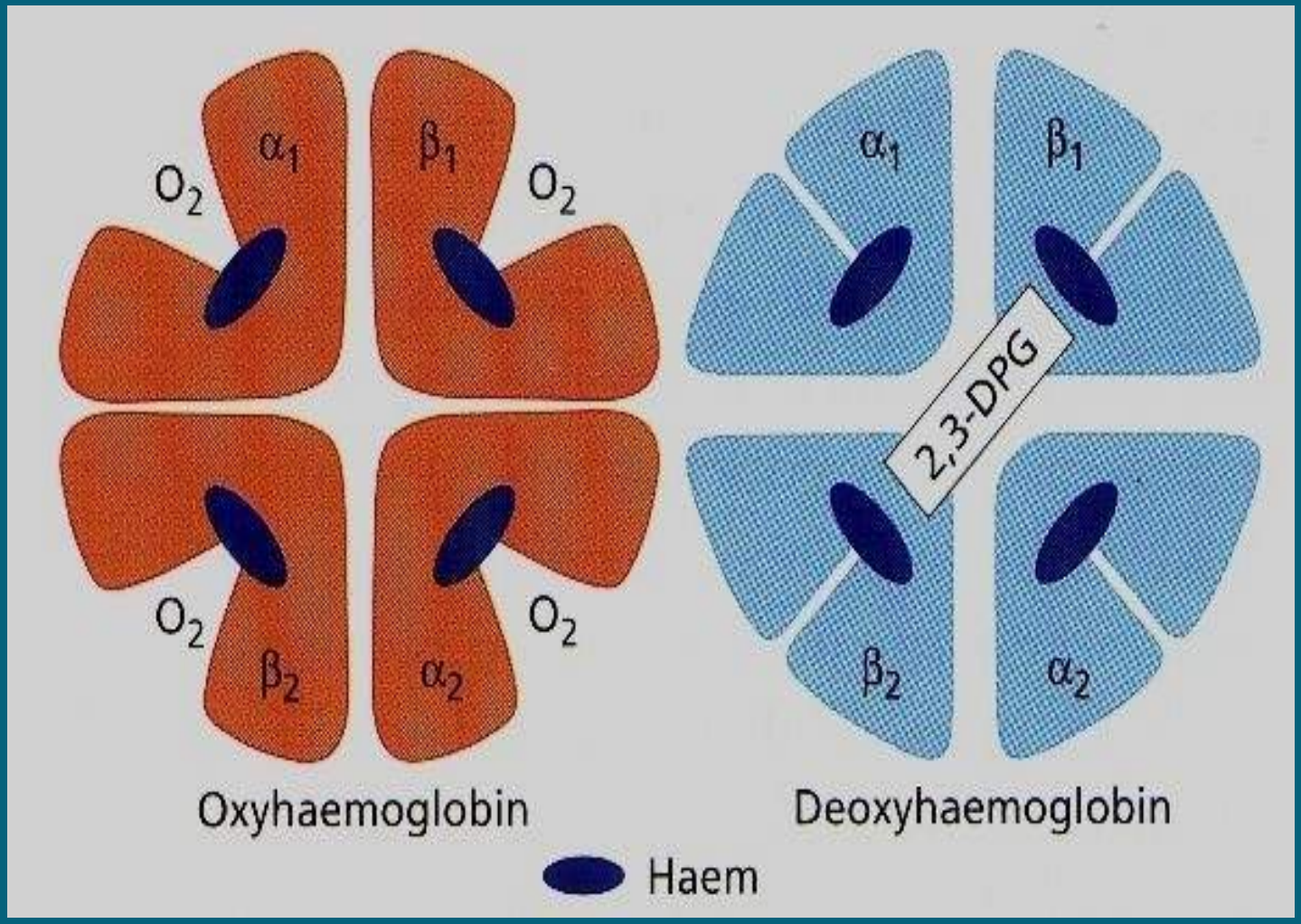
- Hemoglobin SC disease - Another heterozygous form with one sickle gene and another encoding Hemoglobin C.

- The main function of red blood cell
  - Transfer of O_2 from lungs to tissue
  - Transfer of CO_2 from tissue to lungs
- To accomplish this function red cells has haemoglobin (Hb)
- Each red cell has 640 million molecules of Hb

Functions of Haemoglobin

- Oxygen delivery to the tissues
- Reaction of Hb & oxygen
  - Oxygenation not oxidation
  - One Hb can bind to four O_2 molecules
  - Less than .01 sec required for oxygenation
  - β chain move closer when oxygenated
  - When oxygenated 2,3-DPG is pushed out
  - β chains are pulled apart when O_2 is unloaded, permitting entry of 2,3-DPG resulting in lower affinity of O_2

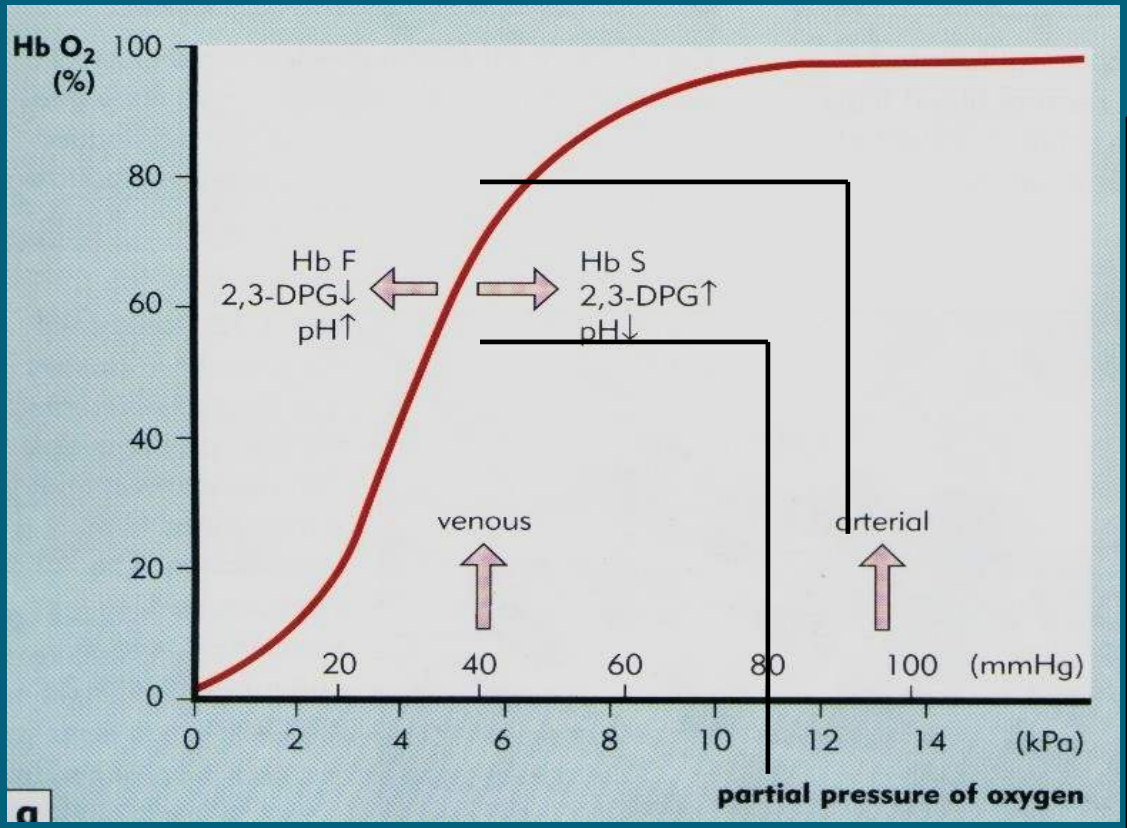
Oxy & deoxyhaemoglobin

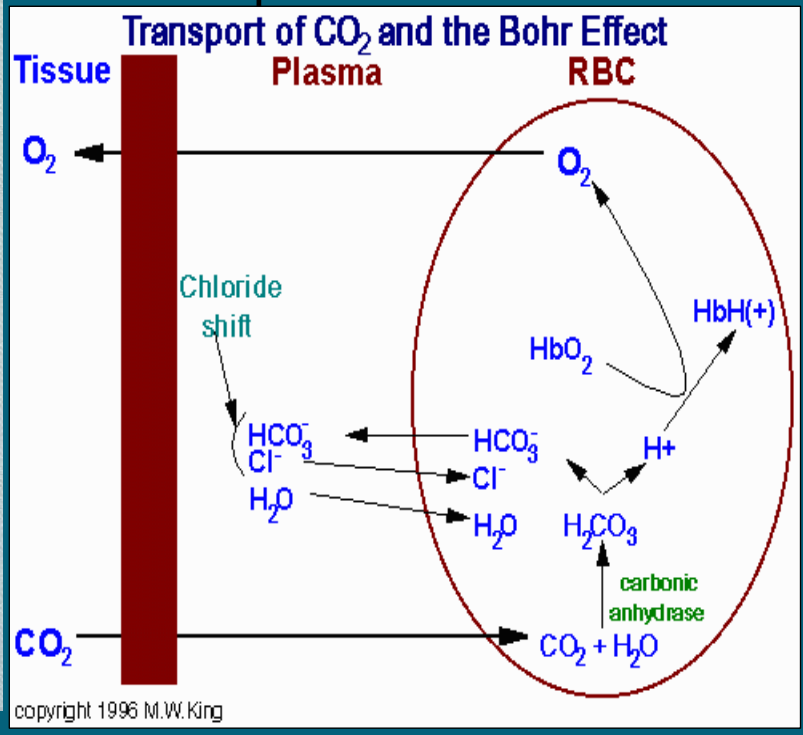


Oxygen-haemoglobin dissociation curve

- O_2 carrying capacity of Hb at different Po_2
- Sigmoid shape
  - Binding of one molecule facilitate the second molecule binding
  - P_{50} (partial pressure of O_2 at which Hb is half saturated with O_2) 26.6mmHg

Hb-oxygen dissociation curve





Hb-oxygen dissociation curve

- The normal position of curve depends on
  - Concentration of 2,3-DPG
  - H^+ ion concentration (pH)
  - CO_2 in red blood cells
  - Structure of Hb

Hb-oxygen dissociation curve

- Right shift (easy oxygen delivery)
  - High 2,3-DPG
  - High H^+
  - High CO_2
  - HbS
- Left shift (give up oxygen less readily)
  - Low 2,3-DPG
  - HbF

- In diabetics whose glucose usually runs high, the percent Hb A_{1c} also runs high. Because of the slow rate of Hb A combination with glucose, the Hb A_{1c} percentage is representative of glucose level in the blood averaged over a longer time (the half-life of red blood cells, which is typically 50–55 days).

- The levels of glycosylated hemoglobin are tested to monitor the long-term control of the chronic disease of type 2 diabetes mellitus (Type2 DM). Poor control of Type2 DM results in high levels of glycosylated hemoglobin in the red blood cells.

- The normal reference range is approximately 4 %–5.9 %. Though difficult to obtain, values less than 7 % are recommended for people with Type 2 DM. Levels greater than 9 % are associated with poor control of diabetes and levels greater than 12 % are associated with very poor control.

- Diabetics who keep their glycosylated hemoglobin levels close to 7 % have a much better chance of avoiding the complications that can sometimes accompany diabetes (than those whose levels are 8 % or higher).

Diagnostic uses

- Hemoglobin concentration measurement is among the most commonly performed blood tests, usually as part of a complete blood count.

Learning Resources

- Lippincott's Biochemistry
- Lecture notes

Introduction

- The **oxygen–hemoglobin dissociation curve** plots the proportion of hemoglobin in its saturated form on the vertical axis against the prevailing oxygen tension on the horizontal axis

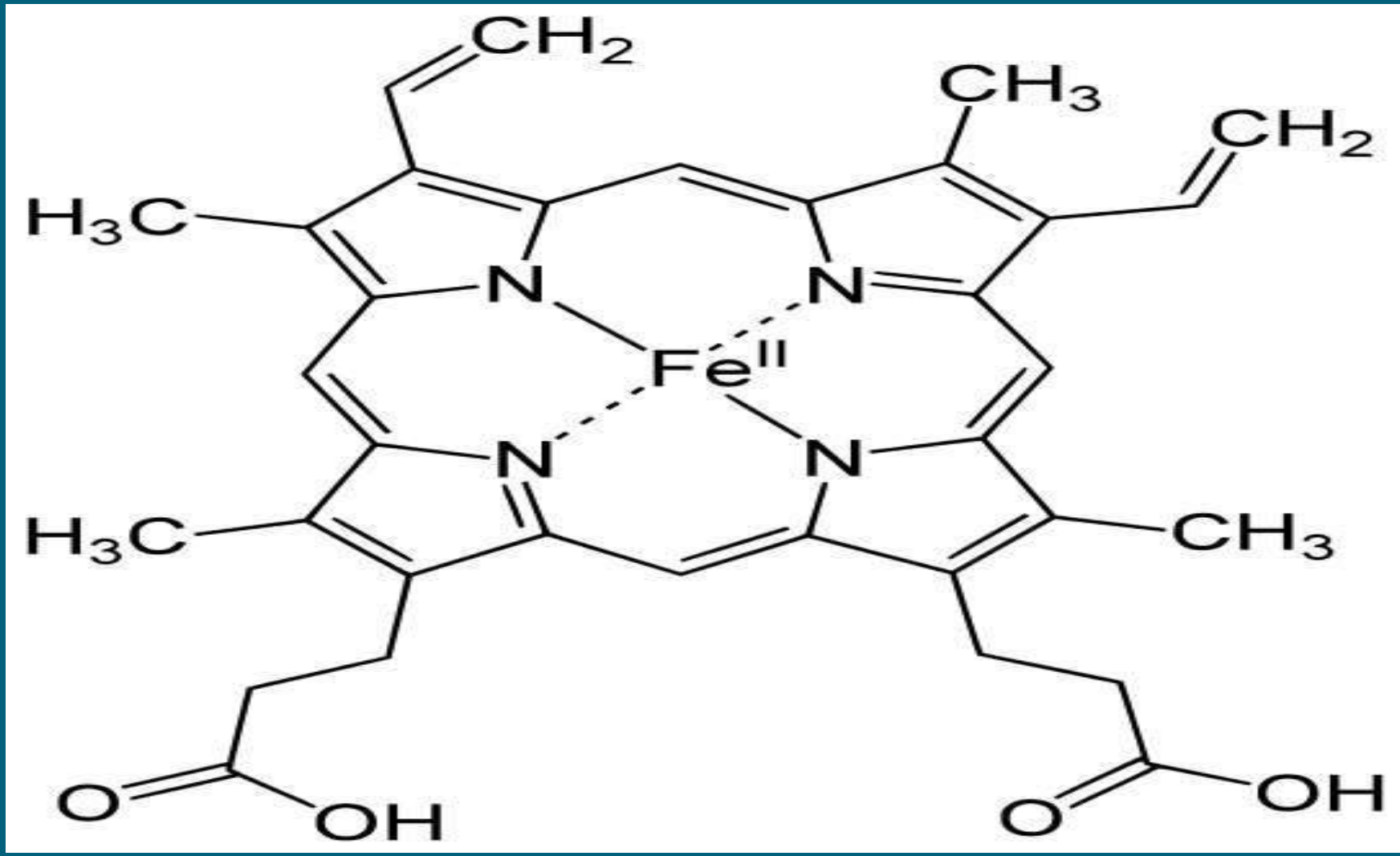
- Important tool for understanding how blood carries and releases oxygen

- More specifically it relates between the
  - Percentage of Oxygen carrying capacity of Haemoglobin and Partial pressure of Oxygen

Hemoglobin

- Heme + Globin – Quarternary
  - Heme: present as 4 subunits connected by polypeptide globin chains as 2 alpha and 2 beta units(adult)
    - Each Heme contains one atom of ferrous iron

Each ferrous iron has ability of binding reversibly to one oxygen molecule (via Oxygenation reaction)


CH2
CH3
CH2
H3C
N
N
Fe II
N
N
H3C
CH3
O
OH
O
OH

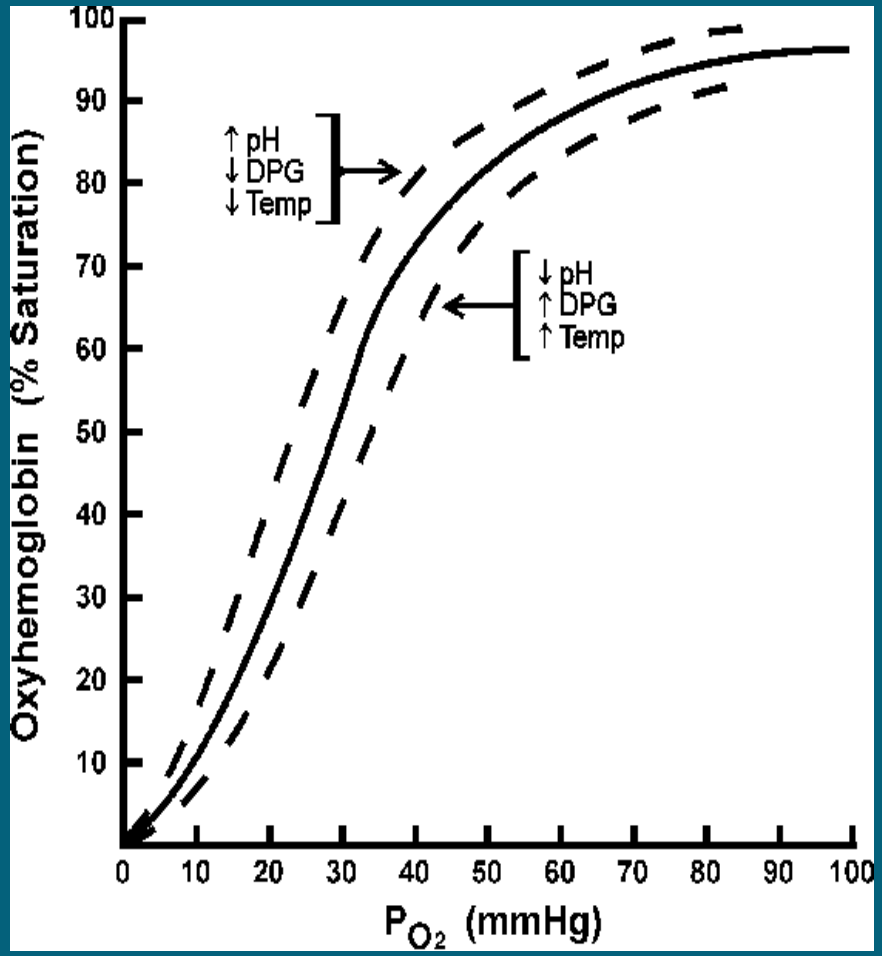
- Hb in 2 forms –deoxyHb and oxyHb
  - In DeoxyHb- globin units are tightly bound(TENSE CONFIGURATION) –which has reduces affinity of Hb molecule to Oxygen

When 1st oxygen is bound- the bonds holding globin chains are released producing RELAXED CONFIGURATION

Which exposes more oxygen binding sites leading to high fold affinity to oxygen binding.

The Curve

- The 'plateau' portion of the curve is the range that exists at the pulmonary capillaries (minimal reduction of oxygen transported until the $p(O_2)$ falls 50 mmHg).
- The 'steep' portion of the curve is the range that exists at the systemic capillaries (a small drop in systemic capillary $p(O_2)$ can result in the release of large amounts of oxygen for the metabolically active cells).



- At pressures above about 60 mmHg, the standard dissociation curve is relatively flat, which means that the oxygen content of the blood does not change significantly even with large increases in the oxygen partial pressure

- Although binding of oxygen to hemoglobin continues to some extent for pressures about 50 mmHg, as oxygen partial pressures decrease in this steep area of the curve,. the oxygen is unloaded to peripheral tissue readily as the hemoglobin's affinity diminishes. The partial pressure of oxygen in the blood at which the hemoglobin is 50% saturated, typically about 26.6 mmHg for a healthy person, is known as the P50

P50

- P50 is a conventional measure of hemoglobin affinity for oxygen

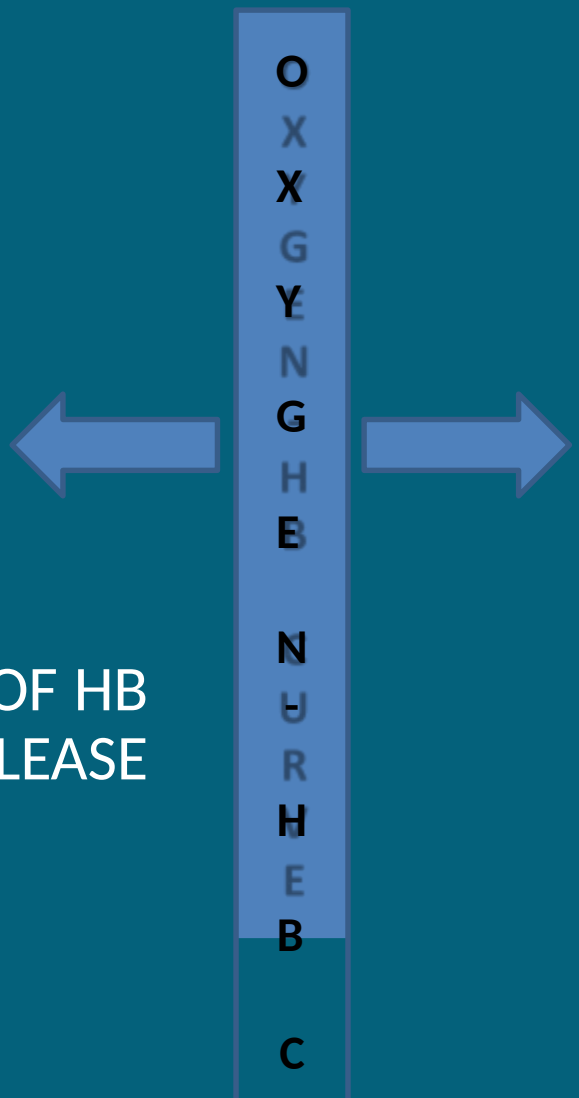
- In the presence of disease or other conditions that change the hemoglobin's oxygen affinity and, consequently, shift the curve to the right or left, the P50 changes accordingly

- Increased P50 indicates a rightward shift of the standard curve, which means that a larger partial pressure is necessary to maintain a 50% oxygen saturation. This indicates a decreased affinity
- Conversely, a lower P50 indicates a leftward shift and a higher affinity.

- SHIFT TO THE LEFT
  - AS IN PULMONARY CAPILLARIES

HIGH pH
DECREASED TEMP.
DECREASED CO2
FETAL HB
METHAEMOGLOBINEMIA

- INCREASED AFFINITY OF HB TO OXYGEN –LESS RELEASE OF OXYGEN



- SHIFT TO THE RIGHT
  - AS IN PLACENTA AND MUSCLES

HIGH Ph
INCREASED TEMP.
INCREASED CO2
INCREASED 2,3 DPG

- DECREASED AFFINITY OF HB TO OXYGEN- MORE RELEASE OF OXYGEN FROM HB